Exam Date & Time: 23-Feb-2026 (01:15 PM - 03:30 PM)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Computer Science and Applications
Bachelor of Science in Information Technology (B. Sc.(IT))
SEMESTER-I
University Theory Examination (Supplementary Exam-1) February 2026

INTRODUCTION TO PROGRAMMING [CTUC101]

Marks: 50 **Duration: 135 mins.**

SECTION-I

Answer all the questions. Section Duration: 20 mins

1 Which is correct with respect to the size of the data types in C?

1) char < int < float 2) int < char < float 3) char > int > float 4) int < chat > float (1)

Course Outcome: "CO2" / Cognitive Level: "Understand"

2 Oval shape indicates _____ in flowchart.

1) Input 2) Output 3) Start/End 4) Condition (1)

Course Outcome: "CO1" / Cognitive Level: "Understand"

3 Portability is one of the characteristics of C programminglanguage.

1) True 2) False (1)

Course Outcome: "CO2" / Cognitive Level: "Understand"

4 What will be the output of the following C code?

```
#include < stdio.h>
void main()
{
int x = 20;
x %= 3;
printf("%d",x);
}
```

(1)

1) 3 2) 2 3) 20 4) 0

Course Outcome: "CO2" / Cognitive Level: "Evaluate"

5 The statement printf("%d", ++5); will print

1) 5 2) 6 3) An error message 4) Garbage Value (1)

Course Outcome: "CO3" / Cognitive Level: "Evaluate"

6 Associativity of an operator is from Left to Right only.

1) True 2) False (1)

Course Outcome: "CO3" / Cognitive Level: "Understand"

7 strcmp() function is used to compare two strings in C.

1) True 2) False

(1)

Course Outcome: "CO4" / Cognitive Level: "Understand"

8 printf() is a predefined Function.

1) True 2) False

(1)

Course Outcome: "CO2" / Cognitive Level: "Remember"

9 In C language, a _____ is a variable which stores the addressof another variable.

1) Constant 2) Data 3) Pointer 4) None of the above

(1)

Course Outcome: "CO5" / Cognitive Level: "Understand"

10 Which is not an advantage of Flowchart?

1) Better communication 2) Property debugging 3) Effective analysis 4) Improper documentation

(1)

Course Outcome: "CO1" / Cognitive Level: "Remember"

SECTION-II

Answer 4 out of 6 questions.

1 List out and explain different properties of algorithm.

(5)

Course Outcome: "CO1" / Cognitive Level: "Remember"

2 Write a C program to print CHARUSAT message.

(5)

Course Outcome: "CO2" / Cognitive Level: "Application"

3 What do you mean by type conversion? Explain its types withsuitable example.

(5)

Course Outcome: "CO3" / Cognitive Level: "Understand"

4 How does do-while loop differ from for loop? Explain with example.

(5)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

5 Write an algorithm to find area of a rectangle.

(5)

Course Outcome: "CO1" / Cognitive Level: "Application"

6 What is variable? Explain with suitable example.

(5)

Course Outcome: "CO2" / Cognitive Level: "Understand"

SECTION-III

Answer 4 out of 6 questions.

1			Evaluate the use of strlwr() and strupr() functions by comparingtheir functionality, advantages, and limitations in C programming. (5) Course Outcome: "CO4" / Cognitive Level: "Evaluate"
2			Describe the procedure to define and call a user-defined functionin C with suitable example. (5) Course Outcome: "CO5" / Cognitive Level: "Understand"
3			What is array? How does it differ from primitive data type?Describe with suitable example. (5) Course Outcome: "CO4" / Cognitive Level: "Analysis"
4			Differentiate generic pointer (void pointer) from null pointer. (5) Course Outcome: "CO5" / Cognitive Level: "Analysis"
5			Apply the concept of pointers to demonstrate how a pointer can bepassed as an argument to a function, with a suitable example. (5) Course Outcome: "CO5" / Cognitive Level: "Application"
6			Apply your understanding of C programming to identify thelimitations of nested functions and explain how these limitationsaffect program development. (5) Course Outcome: "CO2" / Cognitive Level: "Application"

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Event Name	Subject
University Exam Dec-Feb 2026	CTUC101:INTRODUCTION TO PROGRAMMING

#	Course Outcome	Description	Marks
1	CO1	Be able to solve problems using algorithms and flowcharts.	12
2	CO2	Learn about the structure of C program and able to develop simple programs using the C Programming language with data types, variables and constants.	19
3	CO3	Gain the knowledge of different conditional and iterative statements and expressions.	12
4	CO4	Explore the utilization and manipulation of numbers and characters arrays both.	11
5	CO5	Be able to understand user define function. Also, learn Programing-using Array with Pointer and Structure.	16
		Total	**70**

* NA - CO Not Selected

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#	Level	No of Questions	Marks (%)
1	Remember	3	7(10%)

2	Understand	9	21(30%)
3	Application	4	20(28.57%)
4	Analysis	3	15(21.43%)
5	Create	0	0(0%)
6	Evaluate	3	7(10%)
	Total	22	70 (100%)